Econometrics Midterm

R code and output · 기술통계 · 회귀분석 · 회귀선 · 수식과 더미변수 · 가설검정 · 시뮬레이션 · 19 questions

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Exhibit A · Lab2.R — descriptive statistics (W2)

Used in questions 1–3

```r
practice <- read.csv("Lab2_Practice.csv", header=TRUE, na.strings=".")
competence <- practice$COMPETENCE
breaks <- practice$BREAKS
grade <- practice$GRADE
summary(breaks)   # also competence, grade
sum(competence - mean(competence))
sum((competence - mean(competence))^2)
sum((competence - mean(competence))^2)/(length(competence) - 1)
var(competence); sd(competence)
quantile(breaks); IQR(breaks)
boxplot(grade)
```

summary() · n = 27

Variable	Min	Q1	Median	Mean	Q3	Max
competence	1	4	5	5.185	6.5	10
breaks	0	1	2	3.185	4.5	12
grade	36	72	80	77.26	84	97

Lines 6–10 · output

Line	Output	Value
6	sum of deviations	1.78e-15
7	sum of squares	124.0741
8	÷ (27 − 1)	4.7721
9	var() · sd()	4.7721 · 2.1845
10	IQR(breaks) · IQR(grade)	3.5 · 12

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1. [Exhibit A] Line 5 and the summary() table. Compare the mean and the median of breaks and of grade. What does each comparison say about the shape of the distribution, and which centre and spread measures suit a skewed variable?

2. [Exhibit A] Lines 6–9. Explain each output: why line 6 is (almost) 0, why line 7 squares the deviations, and how lines 8–9 turn 124.0741 into the variance and the standard deviation. Why divide by n − 1?

3. [Exhibit A] Lines 10–11. Using the 1.5 × IQR rule, compute the fences for grade and for breaks and name the points that boxplot() draws as outliers.

Exhibit B · regression.R (condensed, line 18 added) — OLS, regression line, inference and prediction (W5)

Used in questions 4–9

```
social <- read.csv("social.csv", header=T)
perform=social$cryp_scoreFINAL; eye=social$RMET_mean
skill=social$Smean_mean; ravens=social$ravens_mean
plot(perform ~ eye)
eye_regr=lm(perform ~ eye)
abline(eye_regr, col="blue")
abline(h=mean(perform), col="red")
abline(v=mean(eye), col="green")
skill_regr=lm(perform ~ skill)
summary(skill_regr)
confint(skill_regr); confint(skill_regr, level=0.99)
cor(perform, skill)^2
mult_lm=lm(perform ~ skill + ravens)
summary(mult_lm)
cor(skill, ravens)
new=data.frame(skill=c(0.5,1.5), ravens=c(1.5,0.5))
predict(mult_lm, new, interval="confidence", level=0.95)
predict(mult_lm, new, interval="prediction", level=0.95)
points <- seq(min(eye)-3, max(eye)+3, length.out=10000)
pred <- predict(eye_regr, data.frame(eye=points), interval="confidence")
polygon(c(rev(points), points), c(rev(pred[ ,3]), pred[ ,2]), col=rgb(0,0,0,0.1),
border=NA)
```

summary(lm()) · slopes · n = 85

Model	Term	β̂	SE	t	p
~ eye	eye	0.1919	0.1077	1.781	0.0785
~ skill	skill	0.2141	0.1072	1.997	0.0491
~ skill + ravens	skill	0.1712	0.1096	1.562	0.122
	ravens	0.1746	0.1096	1.593	0.115

Fit · all intercepts ≈ 0

Model	R^2	adj. R^2	F	df	p
~ eye	0.0368	0.0252	3.172	1, 83	0.0785
~ skill	0.0459	0.0344	3.989	1, 83	0.0491
~ skill + ravens	0.0745	0.0519	3.300	2, 82	0.0418

Other output

Line	Output	Value
7–8	mean(perform), mean(eye)	0.000, 0.000
11	95% CI of skill	[0.0009, 0.4274]
11	99% CI of skill	[−0.0685, 0.4968]
12	cor(perform, skill)^2	0.04585
15	cor(skill, ravens)	0.2458

predict() · lines 17–18

skill, ravens	fit	95% confidence	95% prediction
0.5, 1.5	0.3475	[−0.034, 0.729]	[−1.627, 2.322]
1.5, 0.5	0.3441	[−0.037, 0.726]	[−1.630, 2.318]

4. [Exhibit B] Lines 4–8. What does each line draw? Where do the red and the green lines cross, and why does the blue OLS line always pass through that point?

5. [Exhibit B] Lines 9–12. From the tables, interpret the skill slope (all variables are standardized), show how t is computed, and interpret R^2. Check two identities that hold in simple regression: $R^2 = cor^2$ and $F = t^2$.

6. [Exhibit B] Line 11. The 95% CI of the skill slope is [0.0009, 0.4274] and the 99% CI is [−0.0685, 0.4968], with p = 0.0491. At which levels is H_0: $\beta_{skill} = 0$ rejected, how do the two intervals show it, and what does "95%" mean?

7. [Exhibit B] Lines 13–14. Interpret the skill coefficient in the multiple regression. Then explain how neither slope is significant at 5% on its own (p = 0.122 and 0.115) while F = 3.30 has p = 0.0418.

8. [Exhibit B] Lines 9, 13 and 15. The skill slope is 0.2141 alone and 0.1712 with ravens, and the slope from regressing ravens on skill equals cor(skill, ravens) = 0.2458 here. Show how the numbers fit together and explain the drop.

9. [Exhibit B] Lines 16–21. Interpret the first prediction and its two intervals, explain why the prediction interval is so much wider, and say what lines 19–21 draw around the regression line.

Exhibit C · W6 — formulas, dummy variables and joint tests (df = wage1 n = 526, mroz n = 753)

Used in questions 10–15

```
m1 <- lm(wage ~ female + educ, data = df)
m2 <- lm(wage ~ educ*female, data = df)
m3 <- lm(wage ~ educ + as.factor(region), data = df)
library(car)
linearHypothesis(m2, c("female = 0", "educ:female = 0"))
restricted <- lm(wage ~ educ + exper, data = df)
unrestricted <- lm(wage ~ (educ + exper)*female, data = df)
anova(restricted, unrestricted)
lpm <- lm(inlf ~ nwifeinc + educ + exper + I(exper^2) + age + kidslt6 + kidsge6, data = mroz)
```

m1 and m2 · estimate (SE)

Term	m1	m2
(Intercept)	0.623 (0.673)	0.2005 (0.844)
female	−2.273 (0.279)	−1.199 (1.325)
educ	0.506 (0.050)	0.539 (0.064)
educ:female		−0.086 (0.104)

m3 · region from northcen, south, west · base = east

Term	Estimate	SE	p
educ	0.535	0.053	<0.001
northcen	−0.664	0.426	0.120
south	−0.598	0.397	0.133
west	0.342	0.473	0.469

Lines 5 and 8 · F-tests

Line	Model	Res.Df	RSS	Df	F
5	restricted	524	5980.7		
5	m2	522	5300.2	2	33.51
8	restricted	523	5548.2		
8	unrestricted	520	4737.8	3	29.65

lpm · fitted values −0.345 to 1.127

Term	Estimate	SE	t
educ	0.0380	0.0074	5.15
kidslt6	−0.2618	0.0335	−7.81
age	−0.0161	0.0025	−6.48

10. [Exhibit C] Lines 2, 7 and 9. Write out the terms that `educ*female` and `(educ + exper)*female` put into the model. What does `:` mean, and why is `I(exper^2)` needed instead of `exper^2`?

11. [Exhibit C] Line 1 (m1). Interpret the female coefficient −2.273 (SE 0.279). Which group is the base, and why can't a male dummy be added as well?

12. [Exhibit C] Line 3 (m3). region has four categories. How many dummies does lm() create from as.factor(region), which one is the base, and how is the south coefficient −0.598 (p = 0.133) read?

13. [Exhibit C] Lines 2 and 5 (m2). What is the return to education for men and for women, and is the gap significant? Line 5 tests female = 0 and educ:female = 0 together and gives F = 33.51. Why can it reject when neither t-test does?

14. [Exhibit C] Line 8. Compute F from the RSS values in the table and state the conclusion. What does the test assume about the error variance?

15. [Exhibit C] Line 9. inlf = 1 if a woman is in the labor force. Interpret the educ and kidslt6 coefficients, and name two weaknesses of the LPM that this fit shows.

Exhibit D · simple regression demonstration.R (condensed) — simulation of estimates, tests and intervals (W5)

Used in questions 16–19

```
1  set.seed(4201)
2  slope=0.2; intercept=3; sampleSize=10
3  fertilizer=rnorm(sampleSize,10,2)
4  yield=intercept+slope*fertilizer+rnorm(sampleSize,0,1)
5  fer_regr=lm(yield ~ fertilizer)
6  summary(fer_regr)$coefficients; confint(fer_regr)
7  sampleCnt=10000; estimator=c(); t=c(); pValue=c()
8  for(rowIdx in 1:sampleCnt) {
9    fertilizer=rnorm(sampleSize,10,2)
10   yield=intercept+slope*fertilizer+rnorm(sampleSize,0,1)
11   coef=summary(lm(yield~fertilizer))$coefficients
12   estimator[rowIdx]=coef[2,1]; t[rowIdx]=coef[2,3]; pValue[rowIdx]=coef[2,4]
13 }
14 hist(estimator); mean(estimator); sd(estimator)
15 rejection=ifelse(pValue<=0.05, "Reject", "Fail to Reject")
16 (result=table(rejection))
17 sampleSize=12; slope=2; sampleCnt=20
18 results=data.frame(index=1:sampleCnt)
19 for(rowIdx in 1:sampleCnt) {
20   fertilizer=rnorm(sampleSize,10,2)
21   yield=intercept+slope*fertilizer+rnorm(sampleSize,0,1)
22   interval=confint(lm(yield ~ fertilizer))
23   results[rowIdx,"inside"]=interval["fertilizer","2.5 %"]<=slope &&
   slope<=interval["fertilizer","97.5 %"]
24 }
25 table(results$inside)
```

Line 6 · first sample, n = 10

Term	Estimate	SE	t	p	95% CI
(Intercept)	4.983	1.509	3.302	0.011	[1.503, 8.463]
fertilizer	0.0616	0.139	0.443	0.669	[−0.259, 0.382]

Lines 14–25 · simulation output

Line	Output	Value
14	mean · sd of estimator	0.2027 · 0.190
16	table(rejection)	Fail 8092 · Reject 1908
25	table(results$inside)	TRUE 19 · FALSE 1
25	missed interval (no. 16)	[1.422, 1.979]

16. [Exhibit D] Lines 1–6. What population model do lines 2–4 create, and what does set.seed() guarantee? In the first sample the slope estimate is 0.0616 with p = 0.669 although the true slope is 0.2. Is that a mistake, and what do you conclude from this sample?

17. [Exhibit D] Lines 7–14. What do coef[2,1], coef[2,3] and coef[2,4] pick out? Over 10,000 samples the slope estimates have mean 0.2027 and sd 0.190. What property of OLS does this show, and what does hist(estimator) picture?

18. [Exhibit D] Lines 15–16. The table prints Fail to Reject 8092 and Reject 1908, and the true slope is 0.2, not 0. What does the 19.1% rejection rate measure, what is the other 80.9%, how could the rate be raised, and what would it be if the true slope were 0?

19. [Exhibit D] Lines 17–25. Twenty samples of n = 12 with true slope 2 give 19 intervals that contain 2 and one (no. 16, [1.422, 1.979]) that does not. Is 19 of 20 what you expect, and what does the 95% level actually promise?

Econometrics Midterm — Model answers

1. 기술통계 · W2 · mean vs median
breaks: mean 3.185 > median 2, a few large values pull the mean up, so the distribution is **right-skewed**
grade: mean 77.26 < median 80, the low score 36 pulls the mean down, so it is **left-skewed**
Skewed data: the mean and the SD are pulled by extreme values, so report the **median** and the **IQR** (Q3 − Q1), which use only the order of the data
Key results: right-skewed · left-skewed · median · IQR

2. 기술통계 · Ch.2 · variance by hand
Line 6: deviations from the mean always sum to **0**, and 1.78e-15 is only floating-point rounding, so raw deviations cannot measure spread
Line 7: squaring makes every deviation positive, SS = $\Sigma(x_i - \bar{x})^2$ = **124.0741**
Line 8: s^2 = 124.0741 ÷ (27 − 1) = 124.0741 ÷ **26** = **4.772**, the same as var()
Line 9: s = √4.772 = **2.185**, the same unit as competence
Why n − 1: $\bar{x}$ uses up one **degree of freedom**, and dividing by n − 1 makes s^2 an **unbiased** estimator of σ^2
Key results: 0 · 124.07 · 26 · 4.772 · 2.18 · unbiased

3. 기술통계 · W2 · IQR and boxplot outliers
Rule: a point is an outlier if it lies below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR
grade: IQR = 84 − 72 = 12, lower fence = 72 − 1.5 × 12 = **54** and upper fence = 84 + 18 = 102, so **36** is an outlier
breaks: IQR = 4.5 − 1 = 3.5, upper fence = 4.5 + 1.5 × 3.5 = **9.75**, so **12** is an outlier
Box: the box spans Q1 to Q3, the bold line is the median, the whiskers stop at the last point inside the fences
Key results: 54 · 36 · 9.75 · 12

4. 회귀선 · W5 · plot() and abline()
Line 4: plot(perform ~ eye) is a scatterplot with the **outcome first**, perform on the y-axis and eye on the x-axis
Line 5–6: lm() fits the OLS line and abline() draws it in blue
Lines 7–8: a horizontal line at $\bar{y}$ and a vertical line at $\bar{x}$, which cross at the point of **means** ($\bar{x}$, $\bar{y}$) = (0, 0)
Why: the first-order condition gives $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$, so $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$ and the OLS line **always passes through** ($\bar{x}$, $\bar{y}$)
Here all variables are standardized, so the means are 0 and the **intercept** is ≈ 0
Key results: outcome first · mean · always · intercept

5. 회귀분석 · Ch.2 · summary(lm())
Slope: one standard deviation more social skill goes with **0.214** SD higher team performance on average
t = $\hat{\beta}$ ÷ SE = 0.2141 ÷ 0.1072 = **1.997**, with df = 85 − 2 = 83
R^2 = **0.0459**: skill explains about 4.6% of the variation in performance
Identity 1: in simple regression R^2 = **cor**(perform, skill)^2 = 0.04585, which line 12 prints
Identity 2: the overall F equals $t^2 = 1.997^2$ = **3.989**, with the same p = 0.0491
Low R^2: does not make the slope useless, it can still be a precise ceteris paribus estimate (Ch.3)
Key results: 0.214 · 1.997 · 0.0459 · cor · 3.989

6. 가설검정 · Ch.4 · t-test and confidence interval
5% level: p = 0.049 < **0.05**, so H_0 is rejected, and the 95% CI **excludes 0**
1% level: p > **0.01**, so H_0 is not rejected, and the 99% CI contains 0
Rule: a two-sided test at level α rejects exactly when the 100(1 − α)% CI excludes the null value
95%: in **repeated samples** 95% of intervals built this way cover the true β, any single interval either contains it or not
Key results: 5% · 1% · excludes 0 · repeated samples

7. 가설검정 · Ch.4 · separate t-tests vs the F-test
Skill: holding ravens fixed (**ceteris paribus**), one SD more social skill goes with 0.171 SD higher performance
t-tests: p = **0.122** and 0.115 test one slope at a time, given that the other variable is already in the model
F-test: H_0: $\beta_{skill} = \beta_{ravens} = 0$ against at least one $\neq 0$, F(2, 82) = 3.30 with p = **0.042** < 0.05, so they are **jointly** significant
Why both can hold: skill and ravens share explanatory power (correlation 0.246), so each adds little once the other is in, while together they clearly explain performance
Adjusted R^2: rises from 0.0344 to 0.0519, as expected because ravens has |t| = 1.59 > 1 (Ch.6)
Key results: holding ravens fixed · 0.122 · 0.0418 · jointly

8. 회귀분석 · Ch.3 · omitted variable bias
Formula: simple slope = $\beta_1 + \beta_2 \times \tilde{\delta}_1$ (Ch.3)
Numbers: 0.1712 + **0.1746** × **0.2458** = 0.1712 + 0.0429 = **0.2141**
Why: ravens raises performance ($\beta_2 > 0$) and moves with skill ($\tilde{\delta}_1 > 0$), so without ravens the skill slope picks up part of the ravens effect
Result: the simple slope is biased **upward** by 0.043, the **omitted variable** bias
Note: $\tilde{\delta}_1$ equals the correlation here only because both variables are standardized
Key results: 0.2141 · 0.1746 · 0.2458 · omitted variable · upward

9. 회귀선 · Ch.6 · predict() and the confidence band
Fit: $\hat{y}$ = **0.3475** for skill = 0.5 and ravens = 1.5
Confidence interval: [−0.034, 0.729] covers the **average** performance E(y | x) of all teams with these scores and reflects only the uncertainty of the estimated coefficients
Prediction interval: [−1.627, 2.322] covers **one team**, so it adds that team's own error, $se(\hat{y})^2 + \hat{\sigma}^2$ with $\hat{\sigma}$ = 0.974, and is always **wider**
Lines 19–21: predict() gives the confidence limits at 10,000 eye values and polygon() shades the confidence **band**, which is narrowest at the mean of eye and widens away from it
Key results: 0.3475 · average · one team · wider · band

10. 수식과 더미변수 · W6 · formula operators
Line 2: educ*female = educ + female + **educ:female**, both main effects plus their **interaction**
: alone adds only the product term educ × female
Line 7: (educ + exper)*female = educ + exper + female + educ:female + exper:female, so with the intercept the model has **6** coefficients
Line 9: inside a formula **^** means **crossing** (`exper^2` would just be exper), so **I()** makes R compute the arithmetic square and add exper squared as a regressor
Key results: educ + female + educ:female · interaction · six · I() · crossing

11. 수식과 더미변수 · Ch.7 · dummy coefficient
female: among people with the **same education**, women earn about **$2.27** per hour less than men on average (t = −2.273 ÷ 0.279 = −8.15, p < 0.001)
Base group: men (female = 0), the intercept 0.623 belongs to the men's line and −2.273 shifts it down for women
Male dummy: female + male = 1 for everyone, an exact linear relation with the intercept, the **dummy variable trap** (perfect collinearity, MLR.3)
Caution: this is not proof of discrimination, since female may be correlated with omitted productivity factors such as experience (Ch.7)
Key results: 2.27 · same education · base · dummy variable trap

12. 수식과 더미변수 · Ch.7 · as.factor() with several categories
Dummies: as.factor() makes region categorical, and lm() creates k − 1 = **3** dummies (northcen, south, west)
Base: the first level in alphabetical order, **east**, is left out to avoid the dummy variable trap
south: with the **same education**, workers in the south earn on average $0.60 per hour less than in the east
Significance: p = 0.133 > 0.05, so the difference is **not significant** at 5%, and every region coefficient is a difference from east, not from the overall mean
Key results: three · base · same education · not significant

13. 수식과 더미변수 · Ch.7 · interaction and joint test
Men: return to education = **0.539** dollars per hour per year (female = 0)
Women: 0.539 − 0.086 = **0.453**
Gap: educ:female = −0.086 with p = **0.407**, so no evidence that the return differs
Joint test: F = **33.51** with 2 and 522 df, p < 0.001, rejects that men and women have the same wage line
Why: female and educ:female are almost collinear (correlation 0.96), so their separate t-tests are weak while **jointly** they clearly matter
Key results: 0.539 · 0.453 · 0.407 · 33.5 · jointly

14. 가설검정 · Ch.7 · Chow test by hand
Formula: $F = [(SSR_r - SSR_{ur}) \div q] \div [SSR_{ur} \div (n - k - 1)]$
Numbers: [(5548.2 − 4737.8) ÷ **3**] ÷ (4737.8 ÷ **520**) = 270.1 ÷ 9.111 = **29.65**
Restrictions: q = 3, the female intercept shift and its two slope interactions, and df = 526 − 6 = 520
Conclusion: p < 0.001, so **reject** H_0 that men and women share one wage equation
Assumption: the Chow test needs the **same error variance** in both groups (Ch.7)
Key results: 29.6 · 3 · 520 · reject · same variance

15. 수식과 더미변수 · Ch.7 · linear probability model
educ: one more year of education raises the probability of being in the labor force by 0.038, that is **3.8 percentage points**, ceteris paribus
kidslt6: each child under 6 lowers it by **26.2** percentage points
Weakness 1: fitted values run from −0.345 to 1.127, and 33 of 753 fall **outside** [0, 1]
Weakness 2: Var(u | x) = p(x)[1 − p(x)] depends on x, so the LPM is necessarily **heteroskedastic** and needs robust standard errors
Key results: percentage points · 3.8 · 26.2 · below 0 · heteroskedastic

16. 시뮬레이션 · W5 · the data-generating process
Model: yield = 3 + **0.2** × fertilizer + u, with fertilizer ~ N(10, 2^2) and u ~ N(0, 1), so the true slope is known
set.seed(4201): fixes the random number generator, so every run draws the same samples and the results are **reproducible**
First sample: with n = 10, $\hat{\beta}_1 = 0.0616$ is far from 0.2 only by **sampling variation**, not by a mistake
Test: p = 0.669 > 0.05, so we **fail to reject** H_0: $\beta_1 = 0$ even though it is false, a Type II error, and the 95% CI [−0.259, 0.382] contains both 0 and 0.2
Key results: reproducible · 0.2 · fail to reject · sampling variation

17. 시뮬레이션 · Ch.3 · unbiasedness and the sampling distribution
coef matrix: row 2 is the slope, and columns 1, 3 and 4 are the **slope estimate**, its **t value** and its **p-value** (column 2 is the SE)
Mean: 0.2027 ≈ 0.2, so across repeated samples OLS is right on average, $E(\hat{\beta}_1) = \beta_1$: **unbiasedness** (MLR.1–MLR.4)
hist(estimator): pictures the **sampling distribution** of $\hat{\beta}_1$, centred at 0.2 and roughly normal because the errors are normal (MLR.6)
Spread: sd 0.190 shrinks with a larger n or more variation in fertilizer, since $Var(\hat{\beta}_1) = \sigma^2 \div SST_x$
Key results: slope estimate · t · p-value · unbiased · sampling distribution

18. 시뮬레이션 · Ch.4 · rejection rate = power
Power: H_0 ($\beta_1 = 0$) is false here, so every rejection is correct and 1908 ÷ 10,000 = **19.1%** estimates the **power** of the 5% t-test
Type II: the other 80.9% fail to reject a false H_0, a **Type II** error
Raising power: a **larger sample size** or a larger true slope (the script suggests slope 9 or n = 100) makes the effect easier to detect
True slope 0: H_0 is then true and the rejection rate is about α = **5%**, the Type I error rate
Key results: power · 19.1% · type II · larger n · 5%

19. 시뮬레이션 · Ch.4 · confidence-interval coverage
Expected: 0.95 × 20 = **19** intervals, exactly what this run gives, and interval 16 is the expected one-in-twenty miss
Meaning: 95% is a property of the method, in **repeated samples** 95% of the intervals built this way cover the **true slope**
Single interval: once computed it either contains 2 or it does not, so "a 95% probability that β_1 lies in [1.422, 1.979]" is the wrong reading
Key point: the bounds are **random**, β_1 is fixed (Ch.4)
Key results: 19 · repeated samples · true slope · random